



Muffakham Jah College of Engineering and Technology
Mechanical Engineering Department



PROJECT WORK PRESENTATION

A STUDY ON THE CHANGES IN FLEXURAL PROPERTIES OF E-GLASS/EPOXY LAMINATE COMPOSITE WITH THE INTRODUCTION OF ARTIFICIAL DELAMINATION

PRESENTED BY

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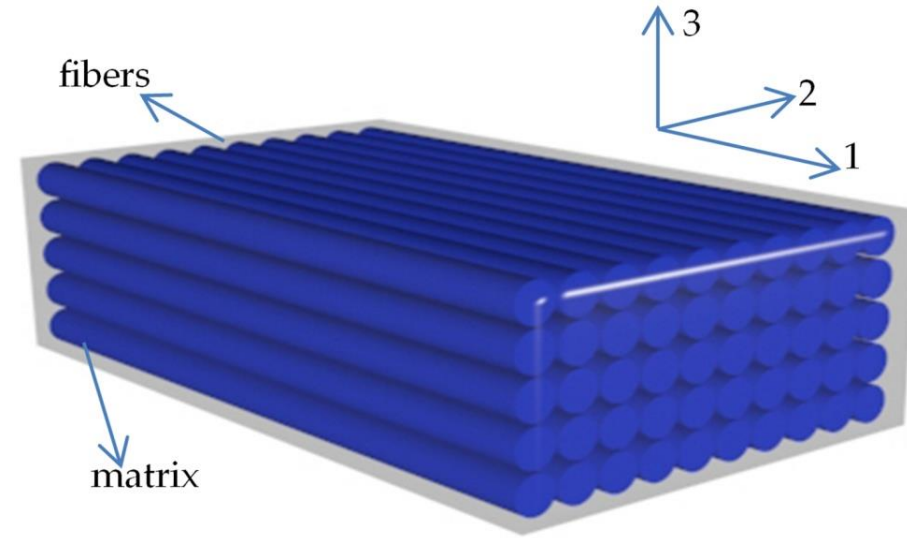
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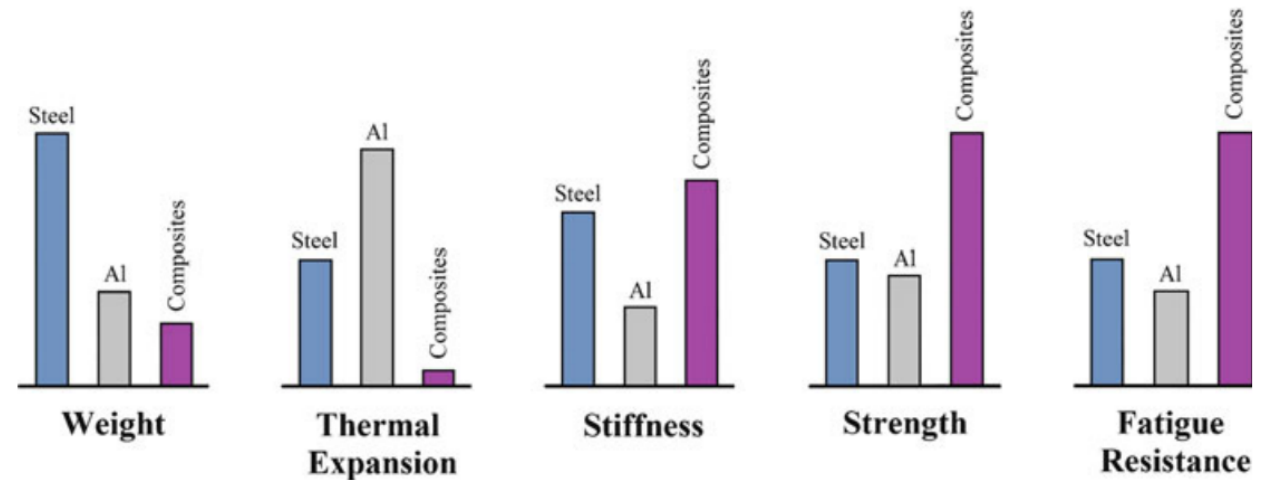
Internal guide: Dr. MOHD. SADAK ALI KHAN, Professor, Mechanical Dept.

Composite materials

Material made from two or more constituent materials with significantly different physical or chemical properties that, when combined, produce a material with **characteristics different from the individual components**.

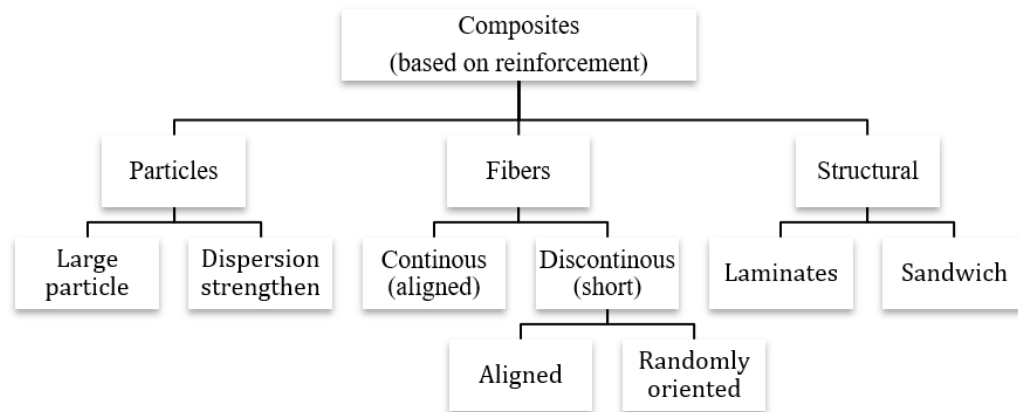


Comparison, for illustrative purposes, between conventional monolithic materials, such as aluminium and steel, and composite materials.

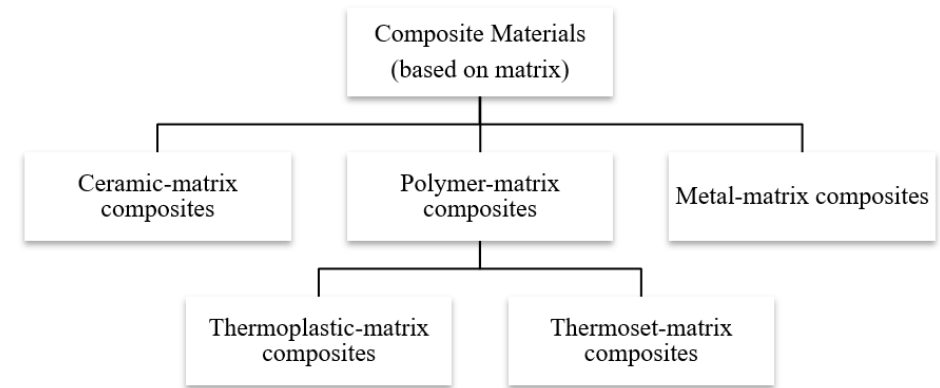


Classification of Composite Materials

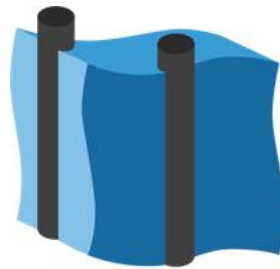
Classification based on the type of reinforcement



Classification based on matrix material



Reinforcement by (a) continuous fibers (b) short fibers or whiskers (c) particulates



(a) **Fibers**



(b) **Whiskers**



(c) **Particulates**

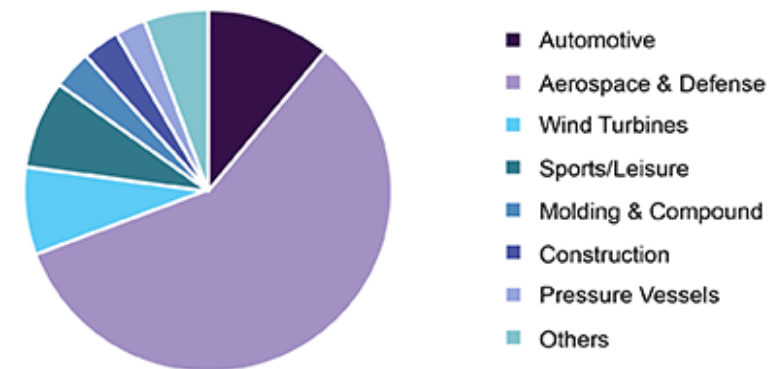
Fiber Reinforced Plastic Composite Materials



Fiber reinforced plastic is that category wherein the mechanical **strength** and **elasticity** of the plastics are enhanced through incorporation of fiber materials.

- These composite materials typically exhibit **low weight** and **high strength**. They are so strong that the automotive and aerospace industries are increasingly interested in using them to replace some of the metals.
- Fiber reinforced plastics can be as strong as some metals but they are much lighter and therefore more **fuel efficient**.

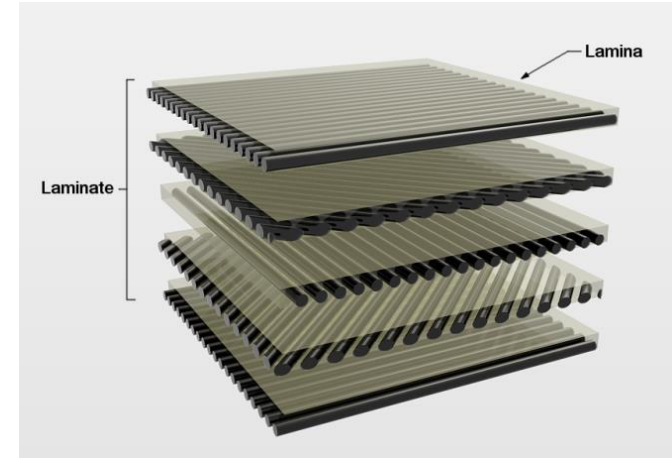
Global carbon fiber reinforced plastic market share, by application, 2017 (%)



Source: www.grandviewresearch.com

Laminates

- A lamina or ply consists of large number of fibers in a thin layer of matrix
- The combinations of these laminas or plies is known as a laminate



Defects in Composite Materials

The following defects arise during a manufacturing process or during the operating life of the component

Debonding

Fracture of Fibers

Matrix Cracking

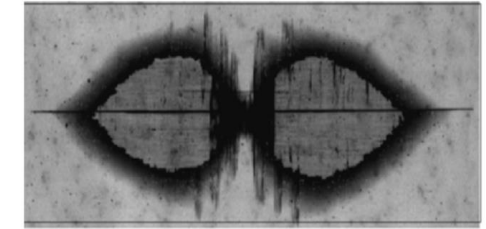
Voids

Fiber Defects

This drastically reduces the performance of the component and eventually leading to failure

Delamination

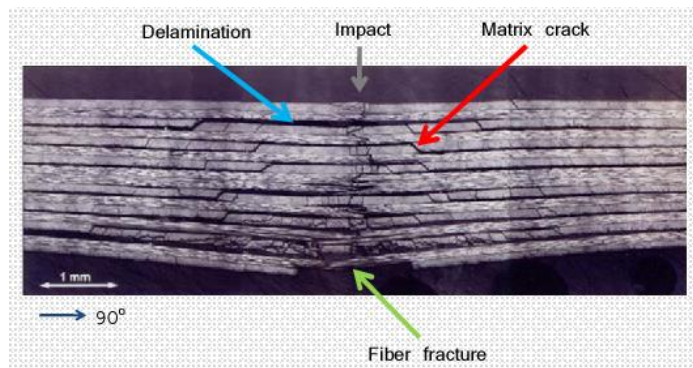
One of the most common failure modes of composite structures is delamination between plies. Delamination is a crack that forms between adjacent plies.



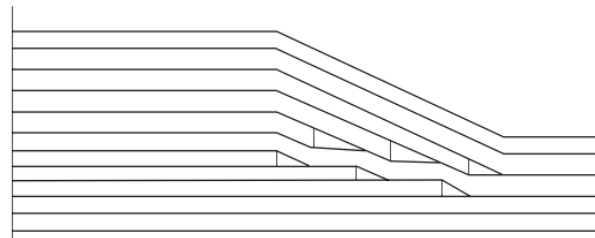
X-Ray image of actual delamination due to impact

Physical phenomena behind delamination onset

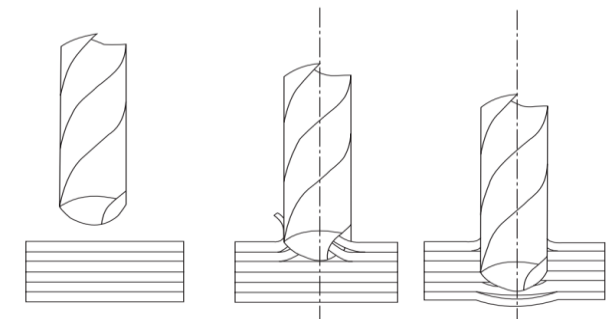
Delamination due to low velocity impact



Tapered laminates



Delamination caused by drilling



Objective of this project

- The bi-woven E-glass/epoxy composite with artificial defect of polytetrafluoroethylene (PTFE) film located at center position between 1st-2nd plies and 5th-6th plies will be manufactured using hand lay-up method. The shape of PTFE film is circular with diameters of 6 mm, 8 mm & 10 mm and square with side length of 6 mm, 8 mm & 10 mm. The composite specimen will consist of twelve layers of glass fabric.
- Experiments will be conducted in flexure with three-point bending test to determine the behavior of laminate.
- The results will be compared to those laminates without artificial defect.

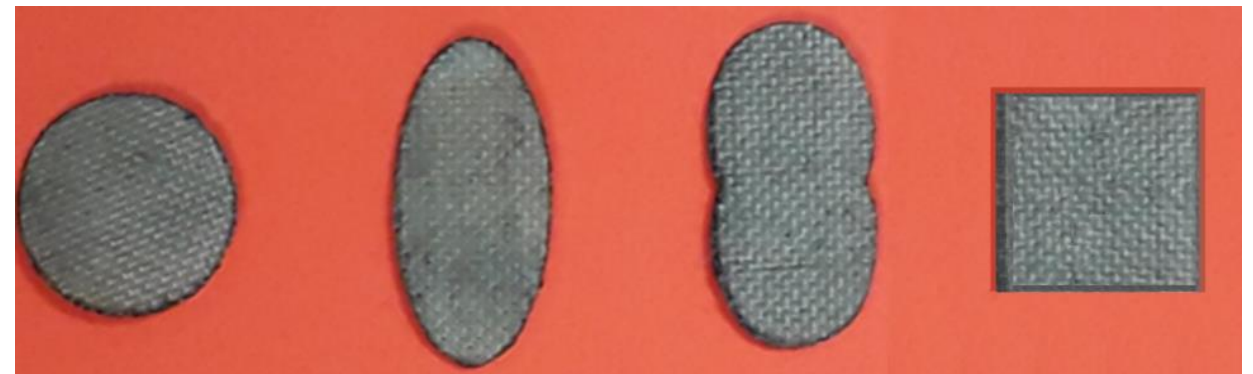
Manufacturing

Material Selection

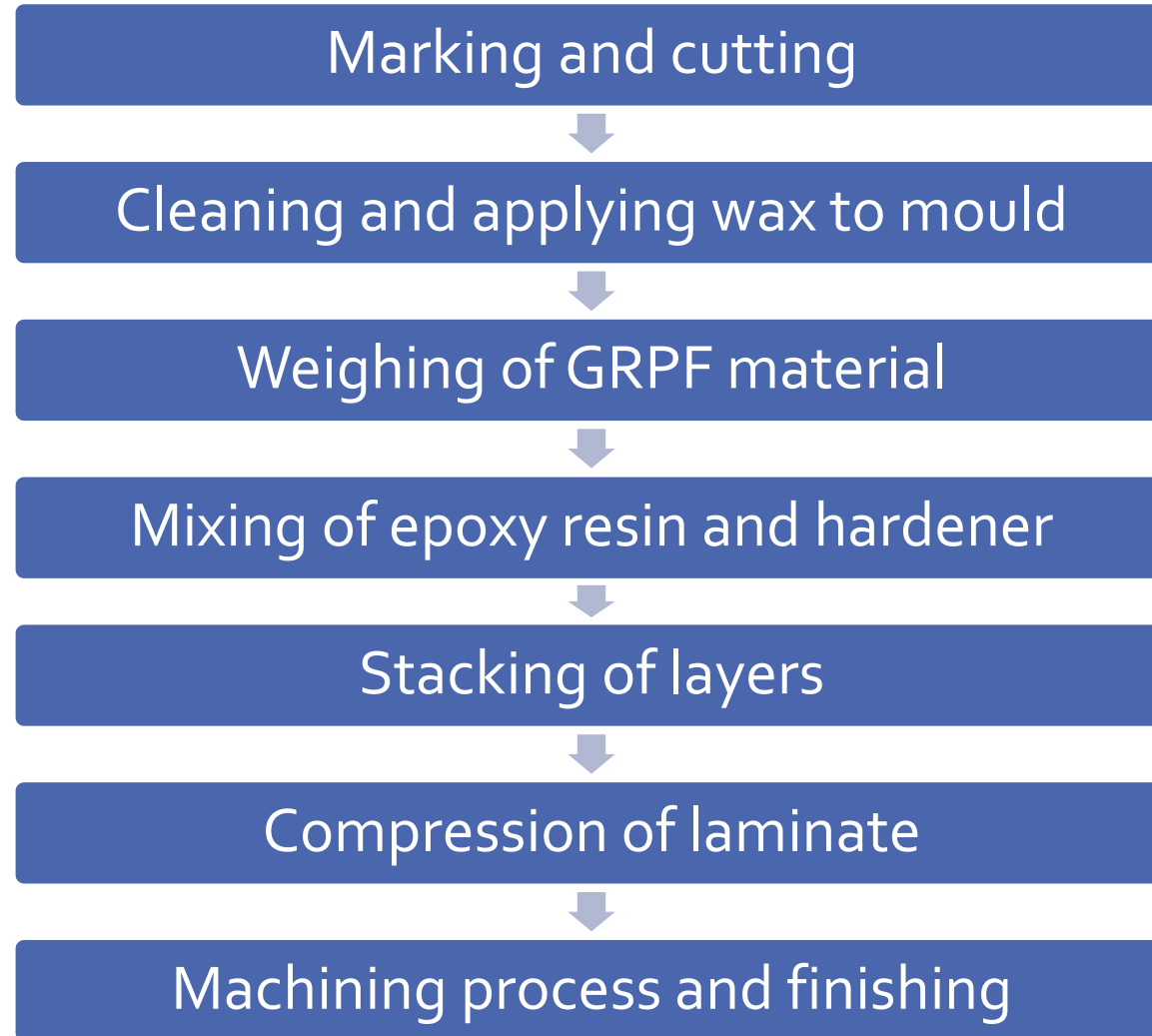
- Plain weave E-glass fabric (7 Mil)
- High strength to weight ratio
- Bi-directional fiber
- Tensile strength 1750MPa
- Matrix is epoxy resin L-12
- Hardener is K-6
- Used for solidification of resin
- Pot life of resin with hardener is 20 mins

PTFE Film

- Polytetrafluoroethylene film used as artificial insert to create delamination
- Circular and Square shaped films were used
- Diameter of circular insert : 6mm, 8mm, 10mm
- Side length of square insert : 6mm, 8mm, 10mm



Laminate Preparation



1. Marking and cutting of fiber

- 12 squares of 35 x 35 cm are marked and cut with scissors



2. Cleaning and application of wax

- EN steel matched die mould of dimensions (390 mm x 340 mm) with spacers which can be used for adjusting height of laminate up to 5 mm
- 2mm spacers were used
- Cleaning of mould is very important to make sure no dust or contamination is present
- Mould is cleaned with acetone
- Wax is applied as a releasing agent



3. Weighing of glass fiber and mixing of epoxy

- Fiber glass is weighed
- Same amount of epoxy resin by weight is poured in a container
- Hardener is added in the ratio of 1 part hardener to 10 parts of epoxy



4. Stacking of layers

- 1 by 1 layers were placed in the mould
- Epoxy is added on each layer and was spread with a roller
- 12 plies were used
- PTFE film was placed between 1-2 and 5-6 layers respectively



5. Compression of laminates and machining of specimen

- The mould is closed and tightened with fasteners
- Left for curing for 24hrs
- After removing the laminate test pieces are machine cut and separated
- Sanded with soft grade emery paper for final dimension of 100 x 15 x 21 mm



Specimen with square defect



Specimen with circular defect



Experimentation

Flexural Test

Three-point bending test

Testing procedure

1. The length, width and thickness of test piece were measured by micrometer
2. The bending fixture was placed on the lower cross head of testing machine
3. The test piece was placed on the rollers of the bending fixture
4. The machine was started and the readings were recorded by the computer
5. The graph was plotted between the load and deflection



Calculation and Results

- The 3 point bending test was carried out on the specimens without defects and with artificial defects
- The maximum stress, maximum strain and the modulus of elasticity in bending was calculated by the following formulae

Maximum stress $\sigma_{\max} = \frac{3P_{\max}L}{2bd^2}$

Maximum strain $\varepsilon_{\max} = \frac{6Dd}{L^2}$

Flexural modulus $E_b = \frac{mL^3}{4bd^3}$

where

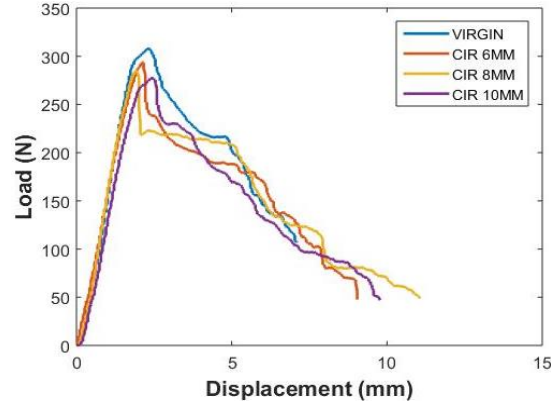
- $P_{\max}$: Maximum bending load (N)
- L: Support span (mm) = 37 mm
- b: Width of the beam (mm) = 15 mm
- d: Depth of the beam (mm) = 2.1 mm
- m: Slope of initial straight portion in load displacement curve (N/mm)
- D: Maximum deflection at the center of beam (mm)

TABLE

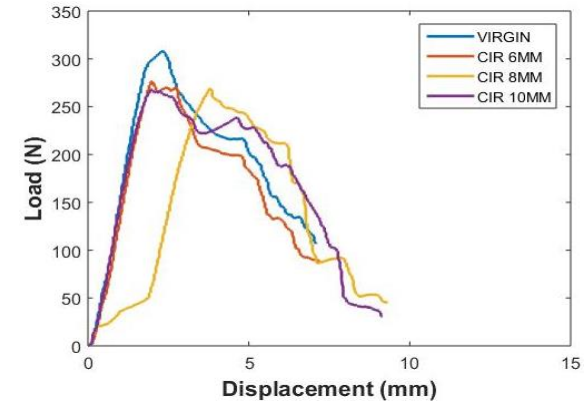
Table 4-1 Three – point bending test results of GFRP virgin and artificial defect specimens

Defect type	Specimen	P _{max} (N)	σ _{max} (N/mm ²)	D (mm)	ε _{max}	m (slope)	E _b (N/mm ²)	% change in E _b
Circular at 1-2 layer	virgin	307.78	258.235	2.32	0.0213528	206.89	18860.3	–
	Ø 6 mm	293.90	246.589	2.11	0.0194200	191.89	17254.7	8.51
	Ø 8 mm	285.37	239.428	1.95	0.0179474	182.02	16379.0	13.15
	Ø 10 mm	277.62	232.926	2.43	0.0223652	170.66	15557.0	17.51
Square at 1-2 layer	virgin	307.78	258.235	2.32	0.0213528	206.89	18860.3	–
	Side 6 mm	267.51	224.448	2.01	0.0184996	167.61	15279.5	18.98
	Side 8 mm	264.77	222.144	2.16	0.0198802	144.26	13150.9	30.27
	Side 10 mm	241.32	202.473	2.38	0.0219050	126.16	11500.8	39.02
Circular at 5-6 layer	virgin	307.78	258.235	2.32	0.0213528	206.89	18860.3	–
	Ø 6 mm	276.15	231.691	2.01	0.0184996	186.25	16738.3	11.25
	Ø 8 mm	269.41	226.042	3.81	0.0350664	174.64	15695.3	16.78
	Ø 10 mm	267.51	224.448	2.01	0.0184996	162.58	14616.9	22.50
Square at 5-6 layer	virgin	307.78	258.235	2.32	0.0213528	206.89	18860.3	–
	Side 6 mm	256.21	214.967	1.90	0.0174872	153.95	14034.3	25.58
	Side 8 mm	241.56	202.670	2.32	0.0213528	135.96	12393.9	34.28
	Side 10 mm	229.35	192.431	2.16	0.0198802	118.88	10837.6	42.53

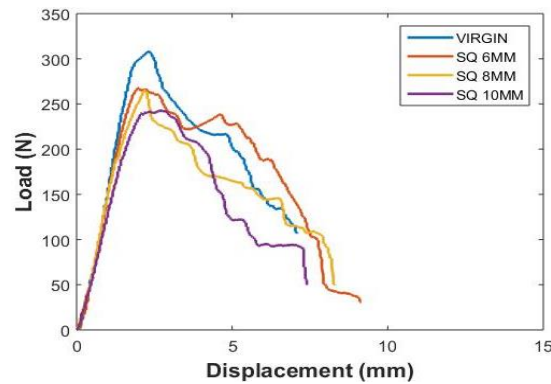
Graphs



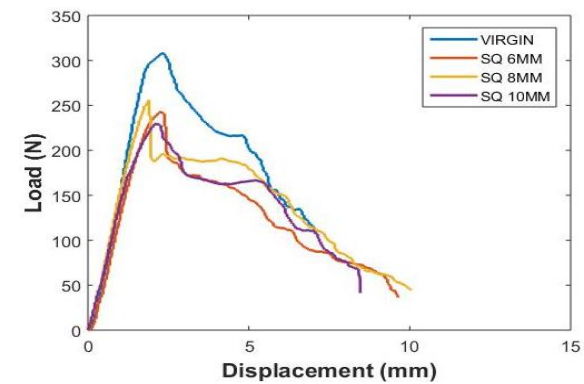
(a)



(b)



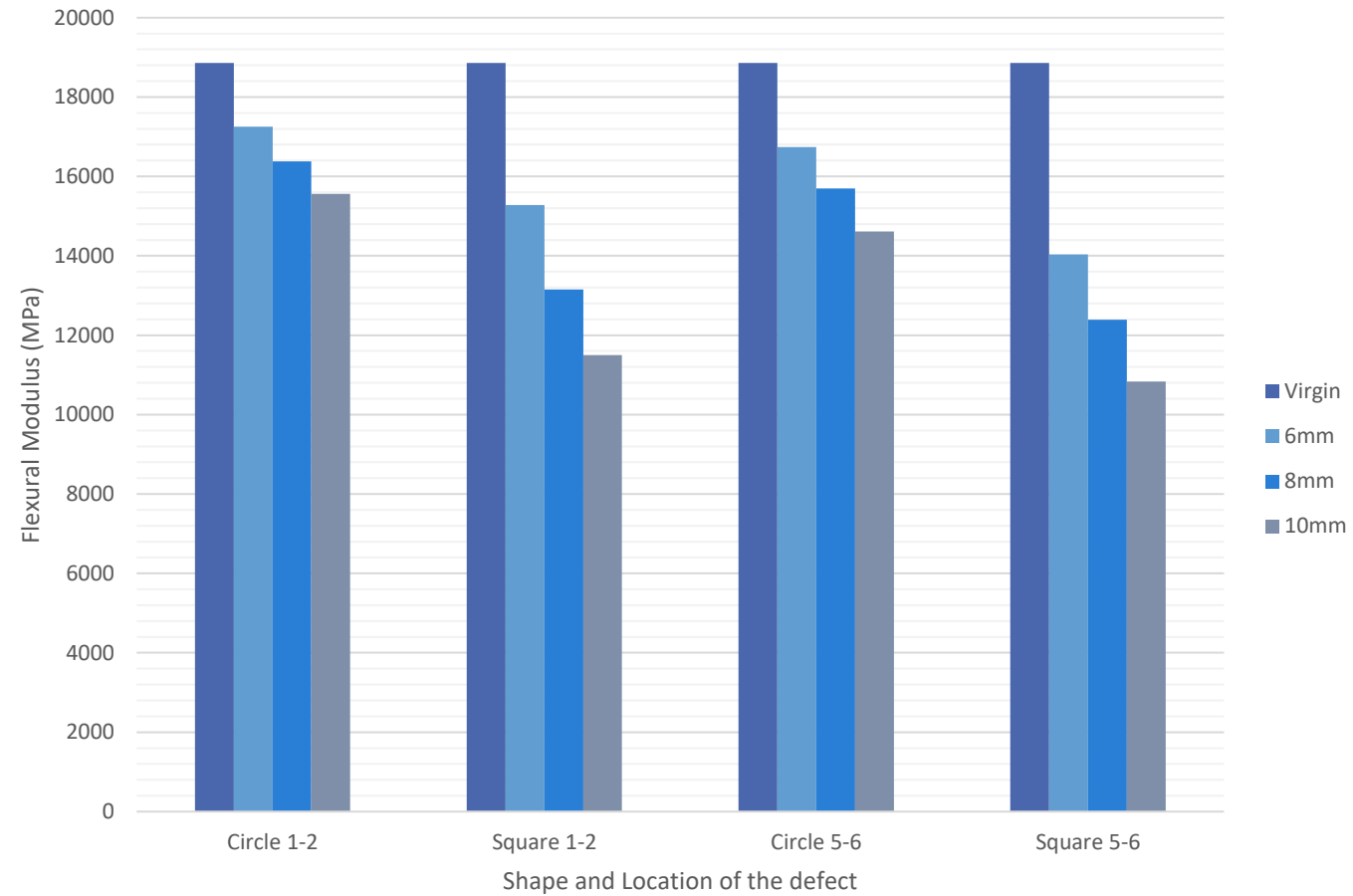
(c)



(d)

Load Displacement curves of samples with
(a) circular defect at 1-2 layer, (b) circular defect at 5-6 layer, (c) square defect at 1-2 layer, (d) square defect at 5-6 layer

Graphs



Variation in flexural modulus with change in position and shape of the defect

Conclusions

The following conclusions can be drawn from the graphs

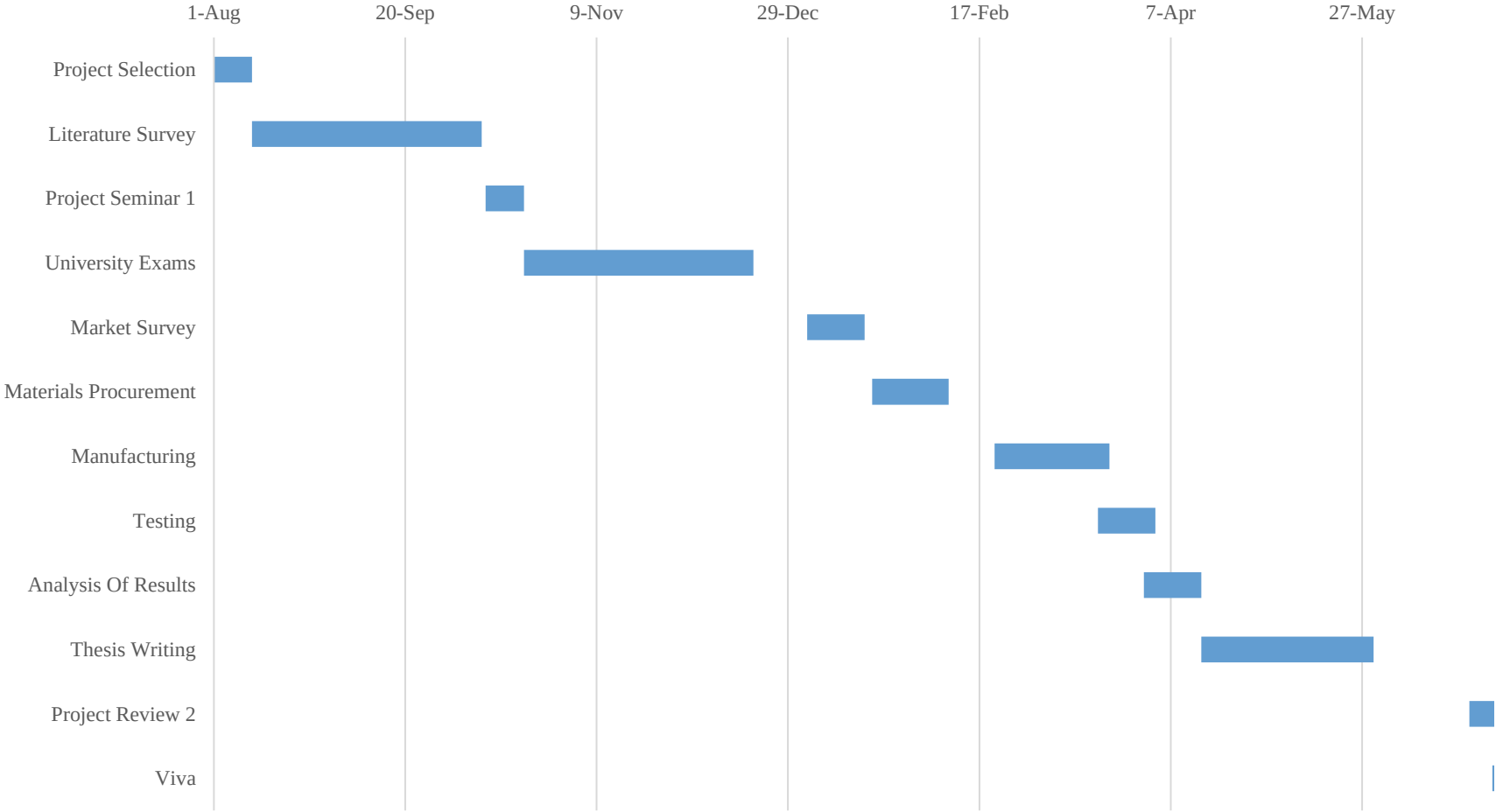
- The mechanical properties of the laminate give a critical response to the presence of the defect, area of the defect and its position in the laminate.
- A maximum drop of **42.5%** in the flexural modulus was observed in the specimen with square defect of side 10 mm at 5-6 layer with respect to the virgin specimen.
- In the Load vs Displacement graphs, **a linear trend is observed** followed by a non-linear behaviour
- The sample with a square defect of side 10 mm has the least modulus irrespective of the position of the defect. However, a **decrease in modulus from 5–8% is observed** when the position of the defect is shifted from the layers 1-2 to 5-6

Future scope:

Considering the work of this project, the following advancements can be made

- Other shapes of the defect which approximate to a **peanut-shape** can be selected
- The mathematical models and CAE models that are generated can be validated

Project Schedule



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Thank you